Shaneeza Hasnani

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Professional Summary

Data scientist completing an M.S. in Business Analytics and Artificial Intelligence (American University, Dec 2026). Project work includes statistical modeling, regression diagnostics, feature engineering, and fraud-risk scoring on 6.3M transactions. Professional experience includes data reconciliation, research automation, and financial-crime analysis.

Skills

Data & Programming: Python (pandas), SQL, R, Snowflake, Git, Excel, Tableau, Power BI, Google Apps Script
Modeling: Regression, classification, decision trees, Random Forest, K-Means, association rules, feature engineering
Statistics: OLS, hypothesis testing, model diagnostics, exploratory data analysis, data validation

Projects

LinkedIn Job Skills & Salary Analytics | *R, K-Means, Apriori, Decision Trees, Regression*

- Integrated job postings, skills, industry, and salary datasets; standardized pay and engineered skill indicators.
- Used Apriori association rules and K-Means/hierarchical clustering to identify recurring skill bundles and distinct job profiles.
- Modeled salary by experience and industry using regression interactions; found that experience-related pay differences varied by industry.

Market Reaction to Corporate Fraud Disclosures | *Python, pandas, statsmodels, SciPy*

- Built a Fama–French three-factor event study of 24 firms using OLS; measured -23.5% cumulative average abnormal return by Day $+10$.
- Estimated expected returns over Days -150 to -30 and abnormal returns around disclosure events.
- Tested significance with parametric and nonparametric methods; checked outliers, alternative windows, and CAPM robustness.

Fraud Detection Pipeline | *Snowflake, SQL, Feature Engineering*

- Built a three-layer SQL pipeline for 6.3M synthetic PaySim transactions; engineered indicators for rule-based fraud scores.
- Engineered sender-balance, recipient-balance, and transaction-type features to score and tier transactions.
- Evaluated four risk tiers; HIGH risk flagged 3,912 fraud cases among 5,145 records (76% precision), vs. 0.13% prevalence.

Spotify Audio Features & Song Popularity Analysis | *R, Multiple Linear Regression, Quarto*

- Cleaned Spotify track data and modeled song energy from audio features using multiple linear regression.
- Grouped track genres and examined loudness, tempo, and other predictors using regression diagnostics.
- Explained about 54% of song-energy variance ($R^2 \approx 0.54$); loudness had the strongest association.

Professional Experience

Schmidt Special Competitive Studies Project | *Research Data Analyst, Arlington, VA* Jan 2026 - May 2026

- Built a research dashboard for 50+ companies using news APIs, financial data, and Airtable; cut meeting prep from 3 hours to 1.
- Reconciled Sheets/Airtable data, integrated Gemini and OpenAI for summaries, and automated action-item reporting.

Guidehouse | *Financial Crime & Investigative Services Intern, NYC* May 2024 - Aug 2024

- Analyzed 20,000+ synthetic wire transfers using Excel risk indicators; identified 10 high-risk patterns.
- Built a Tableau dashboard, presented to 15 stakeholders, and traced transaction flows for investigative analysis.

NYS Attorney General, Medicaid Fraud Control Unit | *Audit Intern, NY* May 2023 - Aug 2023

- Analyzed four Medicaid fraud investigations, contributing to findings of over $500,000 in potentially improper payments.
- Reconciled case records using Excel lookups and pivot tables; identified discrepancies across financial evidence.

Education

American University | M.S., Business Analytics & Artificial Intelligence; Data Science & Statistics concentration Dec 2026
CUNY John Jay College of Criminal Justice | B.S., Fraud Examination and Financial Forensics; CS minor May 2025
Certifications: CFE | UiPath Agentic Automation Developer Associate | Azure AI Fundamentals | SQL Associate